Layer 1 - Data Preprocessing: inputs and verified knowledge

Raw Data Sources
catalogue topic or paid learner request
primary history sources (web)
founder rules in the AvirmathAI skill

Ingestion and Cleaning
lesson schema validation (jsonschema)
source and date cross-check
placeholder binding to verified facts

Mathematics as Code
mathcore module (NumPy, SymPy)
facts dictionary: the only quotable numbers
unit tests bound to every equation

verified facts and lesson inputs

Layer 2 - Orchestration and Reasoning

Lesson Director (Orchestrator)
job state and routing
goal persistence and run trace
bounded retries (max 3)

Eight Specialised Agent Roles
Profiler, Historian, Math Author,
Math Verifier, Lesson Writer,
Animation Director, Visual Critic,
Experiment Builder (plan, act, check)

File-based Memory (no vector database)
procedural: skill files
working: shared artifact store
long-term: project memory across sessions

agent requests: text, speech, frames

Layer 3 - Core Models and Compute

Language Model
frontier LLM via an agent runtime
one model, all eight roles
image input for frame review

Speech Model
Microsoft neural TTS, en-GB Sonia
via edge-tts, word-level timings

Deterministic Renderers and Compute
Manim CE 0.21 + LaTeX frames
FFmpeg assembly, Python 3.13 site generator
local workstation, batched rendering

candidate artifacts

Layer 4 - Governance and Execution Loop

Tool Integration
pytest and SymPy checks
schema validator
snapshot renderer (one still per beat)
browser-based page checks

Verification Gates and Feedback
failed gate returns the exact error
to the responsible agent
strategy recalibration, then retry

Real-time Evaluation Metrics
125 math tests passing
beat overrun = 0 s
frame defects = 0
facts coverage = 100%

gate results and metrics

Layer 5 - Downstream Output and Safety

Safety and Guardrail Boundaries
numbers only from the facts dictionary
history only with sources
fixed branding and voice rules

Observability and Human-in-the-Loop
storyboard approval by the founder
run trace and QA stills
stop after 3 failed retries

Downstream Delivery
narrated video, captions, chapters
lesson web page with assessment
interactive experiment, SEO/AEO/GEO pack
static site deployment

feedback loops: a failed gate returns the error to the agents (Layer 4 to 2); learner feedback and on-demand requests return to the inputs (Layer 5 to 1)